Supporting information

Prediction of Monoclonal Antibodies Pharmacokinetics in Human: Identification of a Reference Neonatal Fc Receptor (FcRn) Binding Affinity Using Physiologically Based Pharmacokinetic (PBPK) Modeling

Salih Benamara[1,3], Erik Sjögren[2], Florence Gattacceca[3], Marylore Chenel[2], Antoine Deslandes[1], Laurent Nguyen[1], *Donato Teutonico[1]

[1]Translational Medicine Unit, Quantitative Pharmacology, Development Pharmacometrics, Sanofi, 94400 Vitry-sur-Seine, France

[2]Pharmetheus AB, Uppsala 753 19, Sweden

[3]COMPutational Pharmacology and Clinical Oncology Department, Inria Sophia Antipolis - Méditerranée, Cancer Research Center of Marseille, Inserm UMR1068, CNRS UMR7258, Aix Marseille University UM105, Marseille, France

* To whom correspondence should be addressed: donato.teutonico@sanofi.com

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Table S1. Estimated drug-specific Kd_{FcRn} values for the 50 mAbs included in the global dataset

drug	M_w (kDa)	source	type of Ig	Kd_{FcRn} (μM)
adalimumab	148	Human	IgG1	1.79
aducanumab	146	Human	IgG1	1.38
atezolizumab	145	Humanized	IgG1	1.80
belimumab	147	Human	IgG1	1.14
bemarituzumab	150	Humanized	IgG2	2.08
benralizumab	150	Humanized	IgG1	1.08
bevacizumab	149	Humanized	IgG1	1.00
bococizumab	145	Humanized	IgG2	1.50
canakinumab	145	Human	IgG1	0.72
daclizumab	144	Humanized	IgG1	0.96
daratumumab	145	Human	IgG1	3.00
dupilumab	147	Human	IgG4	2.00
eculizumab	148	Humanized	IgG2	2.09
emicizumab	146	Humanized	IgG4	1.05
enokizumab	148	Humanized	IgG1	0.81
ensituximab	145	Chimeric	IgG2	2.10
eptinezumab	150	Humanized	IgG1	0.80
erenumab	150	Human	IgG2	1.08
figitumumab	146	Human	IgG2	1.05
fresolimumab	144	Human	IgG4	1.45
fulranumab	145	Human	IgG2	1.15
gedivumab	147	Human	IgG1	1.25
gevokizumab	145	Human	IgG2	0.78

golimumab	150	Human	IgG1	2.24
guselkumab	144	Human	IgG1	1.10
infliximab	144	Chimeric	IgG1	1.30
lecanemab	150	Humanized	IgG1	2.92
lesofavumab	145	Human	IgG1	0.89
mepolizumab	149	Humanized	IgG1	0.85
natalizumab	149	Humanized	IgG4	1.65
obiltoxaximab	148	Chimeric	IgG1	1.10
olokizumab	146	Humanized	IgG1	0.60
omalizumab	149	Humanized	IgG1	0.83
ozanezumab	145	Humanized	IgG1	1.07
pateclizumab	145	Humanized	IgG1	1.60
pertuzumab	148	Humanized	IgG1	0.80
ralpancizumab	145	Humanized	IgG1	1.25
risankizumab	146	Humanized	IgG1	0.72
roledumab	149	Human	IgG1	0.77
sasanlimab	146	Humanized	IgG4	0.86
secukinumab	151	Human	IgG1	0.56
sirukumab	150	Human	IgG1	1.35
spesolimab	146	Humanized	IgG1	0.55
tefibazumab	148	Human	IgG1	0.92
tezepelumab	147	Human	IgG2	0.79
tildrakizumab	147	Humanized	IgG1	0.72
tislelizumab	144	Humanized	IgG1	1.00
tralokinumab	143	Human	IgG4	0.90
trastuzumab	148	Humanized	IgG1	2.00
tremelimumab	149	Human	IgG2	0.80

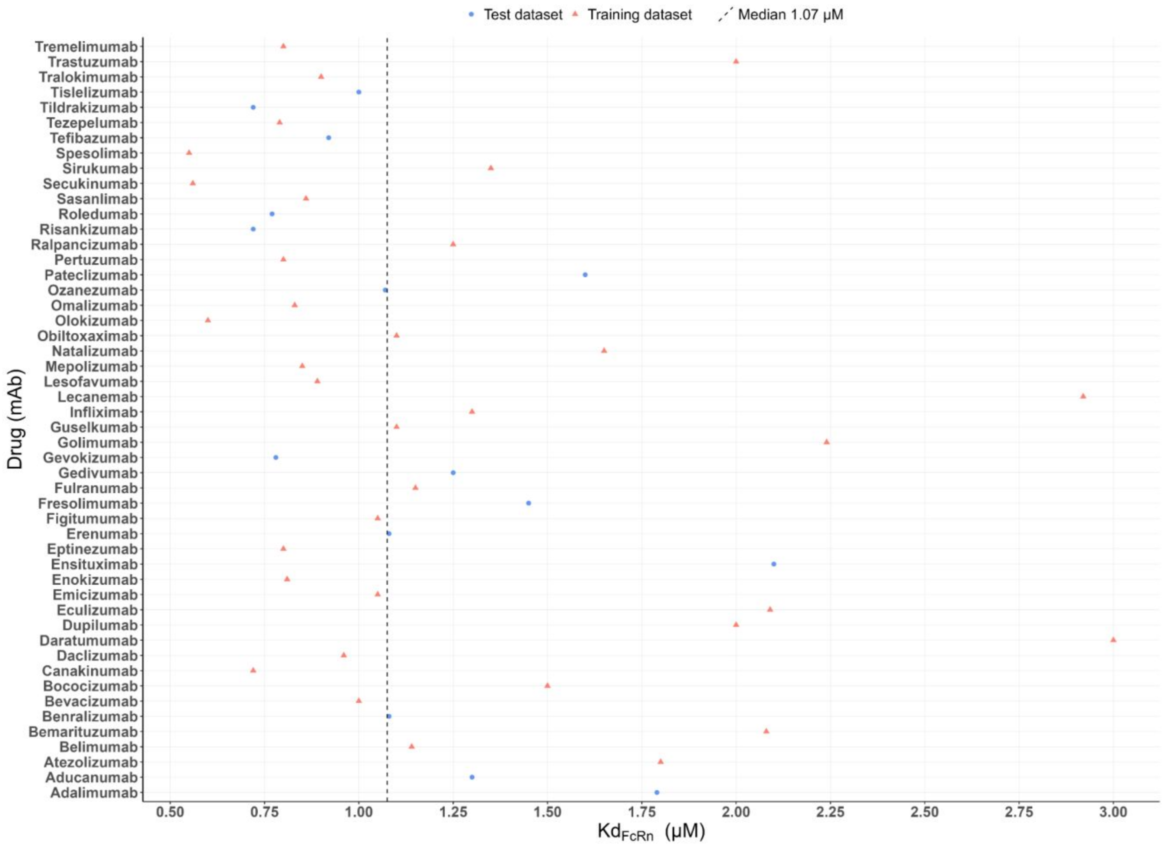

Figure S1. Distribution (colored symbols) and median (dashed line) of estimated drug-specific Kd_{FcRn} values for the 50 mAbs included in the global dataset.

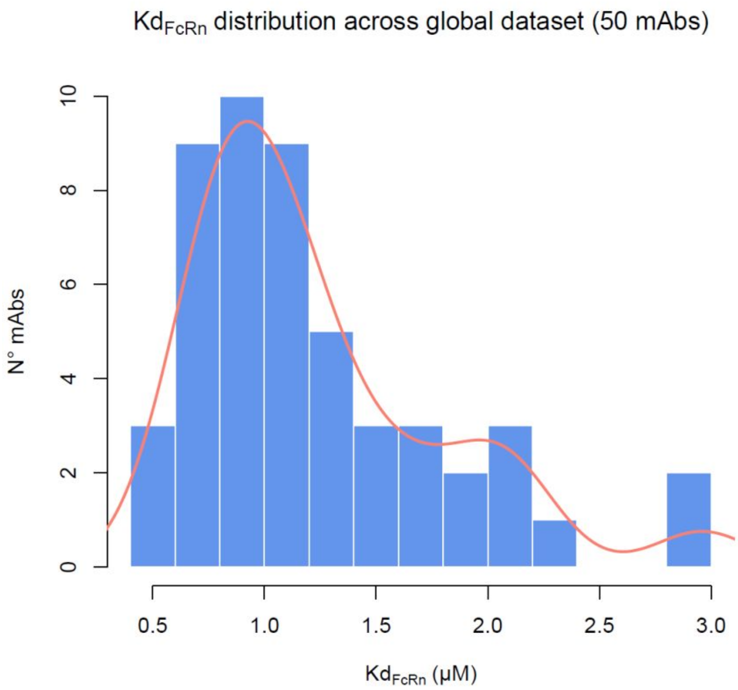

Figure S2. Distribution of Kd_{FcRn} values across 50 mAbs included in the global dataset.

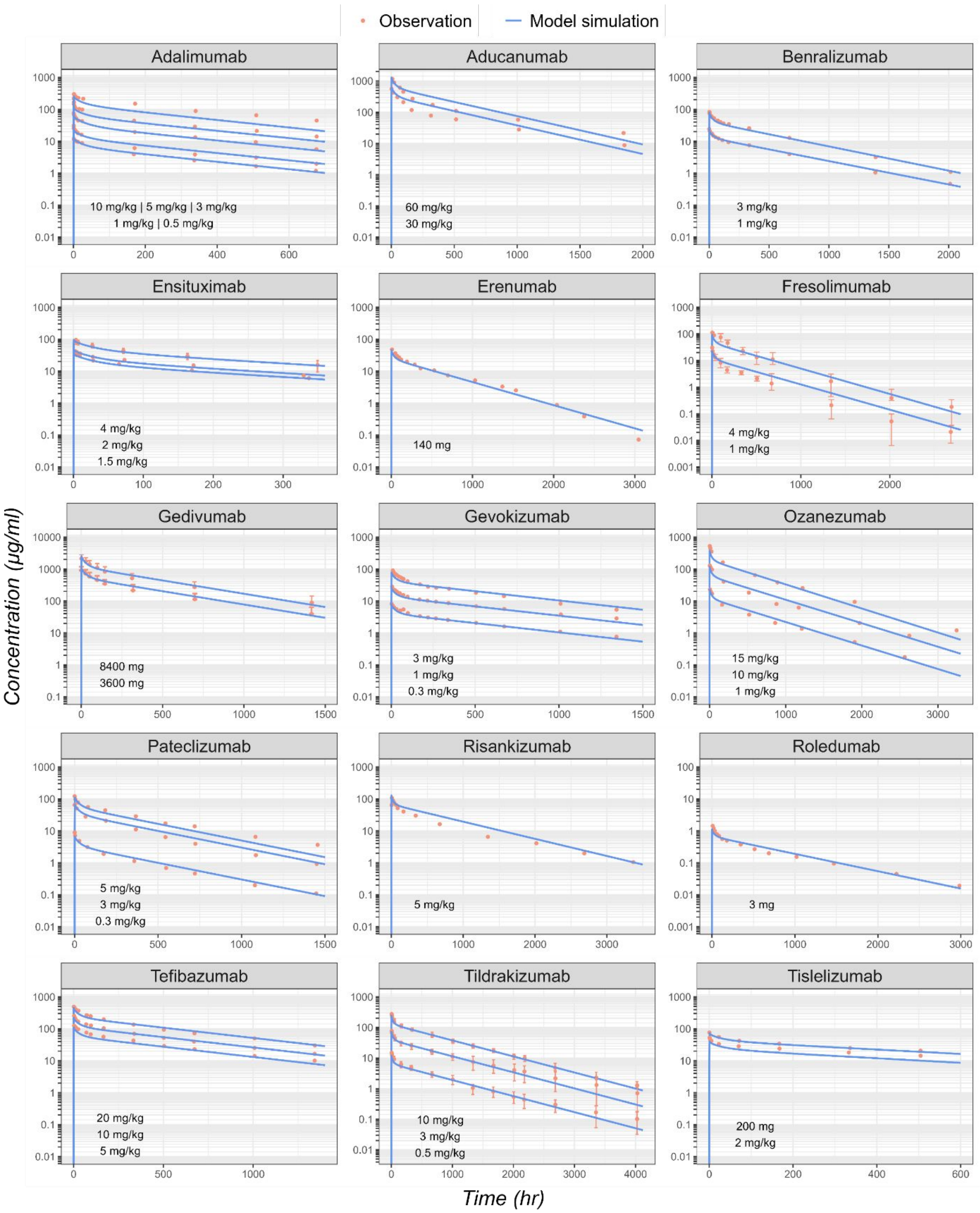

Figure S3. PBPK model simulated (lines) concentration versus time profiles after IV administration compared to observations (dots) for the 15 mAbs included in the test dataset, using individual drug-specific estimated Kd_{FcRn} value for each mAb.

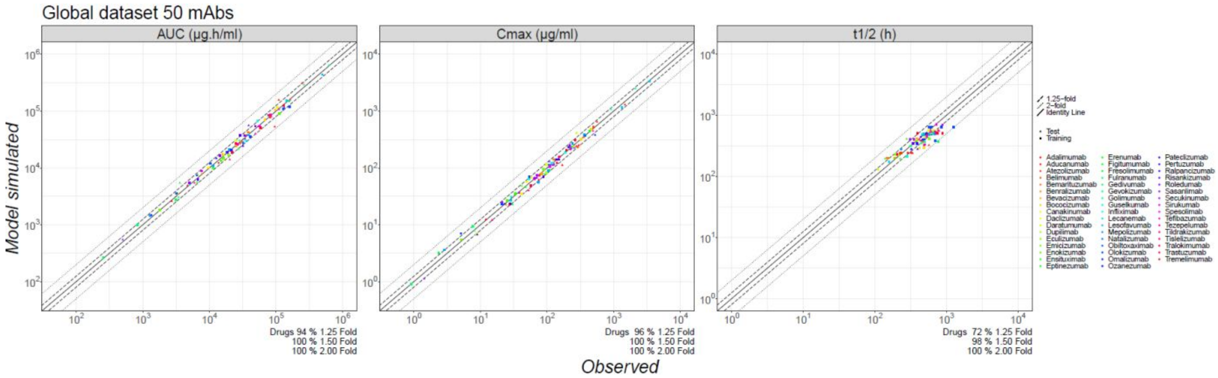

Figure S4. Observed versus PBPK model simulated AUC, Cmax, and $t_{1/2}$ for all the 50 mAbs included in the global dataset. Squares represent training dataset mAbs, and triangles represent test dataset mAbs.

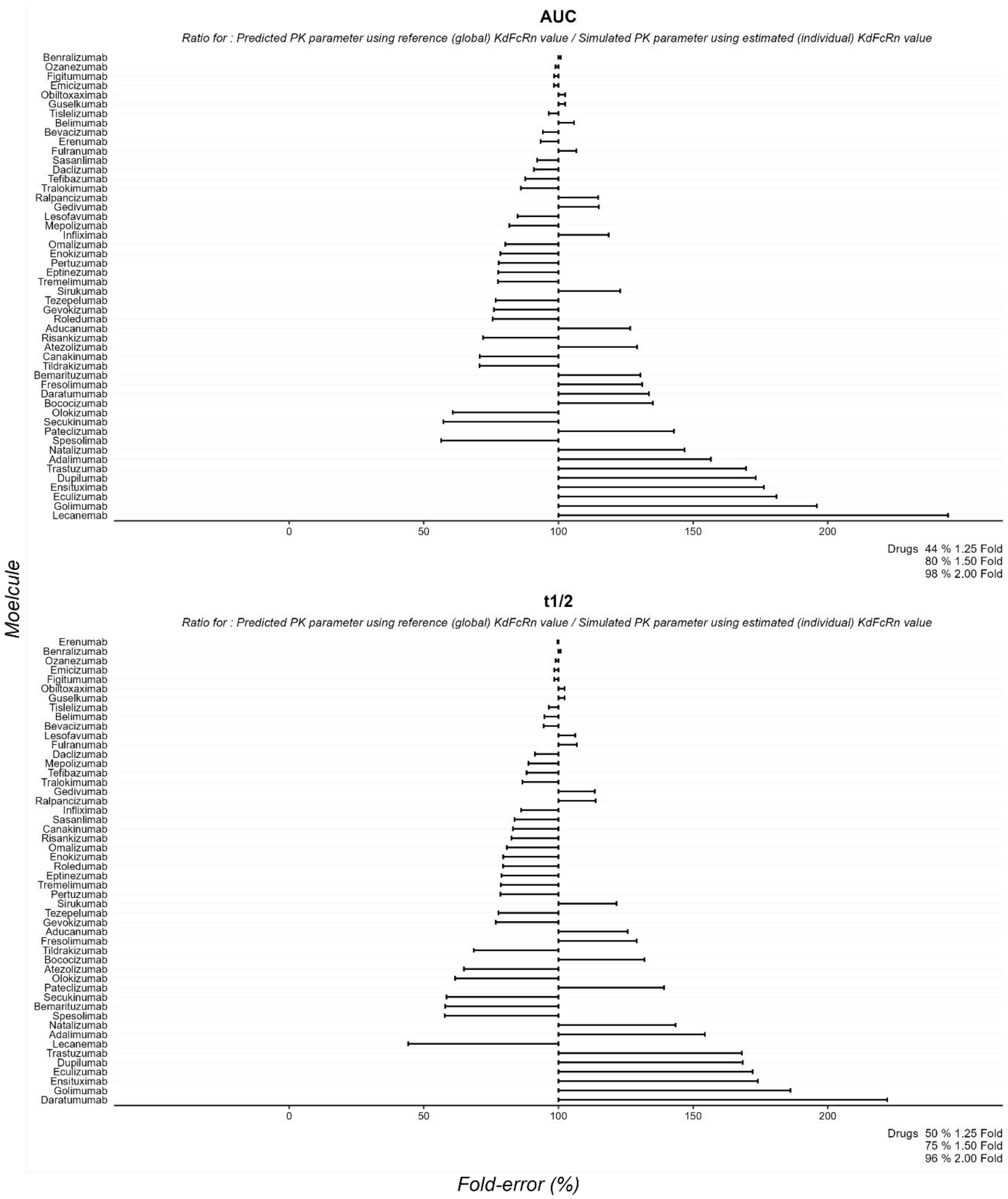

Figure S5. Predicted versus simulated AUC, and $t_{1/2}$ for the 50 mAbs included in the global dataset, using reference $K\mathrm{d}_{\mathrm{FcRn}}$ value of 1.07 µM versus individual drug-specific $K\mathrm{d}_{\mathrm{FcRn}}$ value. Each bar illustrates the percent deviation from unity (100%) of the ratio between predicted PK parameter using the reference (global) $K\mathrm{d}_{\mathrm{FcRn}}$ value and simulated PK parameters using estimated (individual) $K\mathrm{d}_{\mathrm{FcRn}}$ value.